Name: | **Roll No.:**

Mid Term Examination

Math-IV (Sec-A)

IIIT Delhi

1st March, 2026

Full Marks: 60 **Duration: 1 Hour and 30 Minutes**

Instructions:

- *Attempt all questions.*
- **Write your name and roll number on both the question paper and the answer sheet.**
- Write all *intermediate steps* clearly.
- Marks for *proof-type questions* will depend on the *logical progression* of the steps.
- You may quote (without proof) any *proposition or theorem* covered in the lectures and tutorials, but it must be clearly identified.
- Write *clearly* and identify each part of your answers for the benefit of graders.
- The possession of **electronic devices such as mobile phones, laptops, smartwatches etc** and **discussions** are strictly prohibited.

Problems

Problem-1 **[8 marks]**

Let $y(x)$ be the solution of the differential equation

$$xy^2y' + y^3 = \frac{\sin x}{x}, \qquad x > 0,$$

satisfying

$$y\left(\frac{\pi}{2}\right) = 0.$$

(a) Find $y(x)$.

(b) Find the value of $y\left(\frac{5\pi}{2}\right)$.

Problem-2 **[8 marks]**

Let $D = \{(x,y) \in \mathbb{R}^2 : -1 \leq x \leq 1,\ -1 \leq y \leq 1\}$ and define $f : D \to \mathbb{R}$ by

$$f(x,y) = 1 + \sqrt{\hat{y}}, \qquad \text{where } \hat{y} = \max\{y, 0\}.$$

Consider the initial value problem (I.V.P.)

$$\frac{dy}{dx} = f(x,y), \qquad y(0) = 0.$$

(1) Check whether f satisfies Lipschitz condition with respect to y on D.

(2) Determine whether the I.V.P. has

(a) no solution, or

(b) at least one solution.

Justify your answer.

Problem-3 **[8 marks]**

Consider the differential equation

$$M(x,y)\,dx + N(x,y)\,dy = 0,$$

where M and N have continuous first partial derivatives at all points (x,y) in a rectangular domain D.
If the differential equation is exact in D, then show that

$$\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}.$$

Problem-4 **[8 marks]**

Consider the system of linear differential equations

$$\frac{dy_1}{dx} = 5y_1 - 2y_2, \qquad \frac{dy_2}{dx} = 4y_1 - y_2,$$

with the initial conditions

$$y_1(0) = 0, \qquad y_2(0) = 1.$$

(1) Find the solution for the given system of equation.

(2) Find the critical point of the system.

(3) Analyze the nature and stability of the critical point.

Problem-5 **[10 marks]**

Let f_1 and f_2 be two solutions of the second order homogeneous linear differential equation

$$y'' + a_1(x)y' + a_2(x)y = 0.$$

Show that if f_1 and f_2 are linearly independent on $a \leq x \leq b$ and are such that

$$f_1''(x_0) = f_2''(x_0) = 0$$

at some x_0 of this interval, then

$$a_1(x_0) = a_2(x_0) = 0.$$

Problem-6 **[8 marks]**

Let $\phi : (-1,\infty) \to (0,\infty)$ be the solution of the differential equation

$$\frac{dy}{dx} - 2ye^x = 2e^x\sqrt{y},$$

satisfying $\phi(0) = 1$. Then find ϕ.

Problem 7 **[10 marks]**

Each of the two tanks contains 200 gal of water, in which initially 100 lb (Tank T_1) and 200 lb (Tank T_2) of fertilizer are dissolved. The inflow circulation, and outflow are shown in the figure below. The mixture is kept uniform by stirring. Find the fertilizer contents $y_1(t)$ in T_1 and $y_2(t)$ in T_2.